

Loho Christopher Dondo

Mathematics Lecturer • Software Engineer • Cybersecurity Researcher

Ph.D. Candidate | Applied Mathematics

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Professional Profile

Interdisciplinary expert bridging pure mathematics, full-stack engineering, and cybersecurity. Currently pursuing Ph.D. in Applied Stochastic Processes while actively lecturing at university level. Published researcher in reliability engineering and AI-driven security systems.

Education

Bayero University Kano <i>Ph.D. in Applied Mathematics (In Progress),</i> Research Focus: Applied Stochastic Processes, Renewal-Reward Theory, and Reliability Engineering for Critical Infrastructure. Key Areas: Delay Time Analysis, Weibull Modeling, Infrastructure Risk Assessment	Kano, Nigeria 2025–Present
Bayero University Kano <i>M.Sc. in Applied Mathematics</i> Specialization: Stochastic Modeling and Delay Time Analysis	Kano, Nigeria 2021–2024
University of Agriculture Makurdi <i>B.Sc(Ed). Mathematics/Computer Science, Second Class Upper (Honours)</i> Core Areas: Mathematical Modeling, Algorithm Design, Data Structures	Benue, Nigeria 2014–2018

Professional Experience

Skyline University Nigeria <i>Lecturer of Mathematics</i> - Deliver advanced courses: Calculus I–III, Differential Equations, Linear Algebra, and Mathematics for Economists (avg. class size: 45 students) - Supervise undergraduate research projects, with 2 students pursuing graduate studies in computational mathematics - Integrate computational tools (MATLAB, Python, R) into mathematical modeling coursework, improving student engagement by 75% - Course coordinator for Calculus II, developing assessments and maintaining grade consistency across sections	Kano, Nigeria Feb 2025–Present
ASHA International Model School <i>Mathematics & Web Development Instructor</i> - Designed integrated curriculum combining secondary mathematics (SSS 1–3) with introductory web development (HTML, CSS, JavaScript) - Achieved 78% improvement in mathematics pass rates through computational thinking integration - Led project-based learning: 45+ students built functional math tools and interactive quizzes - Mentored technology-career pathway students; 12 admitted to computer science programs (2023–2024) - Introduced coding clubs and annual "Math + Code" hackathons for 80+ participants	Kano, Nigeria 2021–2025
Holy Cross Catholic School <i>Vice Principal of Academics & Mathematics Teacher</i>	Badawa, Kano 2016–2021

- Directed academic operations for K-12 institution (600+ students), managing 25+ teachers and 15 support staff
- Increased WAEC mathematics credit pass rate from 51% to 85% (2016–2021) via data-driven interventions
- Implemented teacher mentorship program reducing teachers turnover by 40% within 18 months
- Redesigned examination system, reducing grading discrepancies by 60% through standardized rubrics
- Chaired Academic Committee; led curriculum reform aligning NERDC standards with 21st-century skills
- Secured 3 grants totaling ₦2.5M for STEM lab equipment and teacher training workshops

Core Competencies

Mathematics & Statistics: Stochastic Processes, Renewal-Reward Theory, Reliability Engineering, Delay Time Analysis, Weibull Modeling, Markov Chains, Queueing Theory

Software Engineering: React, Node.js, TypeScript, Python, MATLAB, R, Tailwind CSS, Vite, Git, Vercel, AWS (basic)

Cybersecurity & AI: Digital Footprint Analysis, AI-driven Threat Triage, Privacy-first Architecture, Threat Modeling, Risk Assessment Frameworks

Academic & Research: LaTeX, Scientific Writing, Peer Review, Grant Writing, Conference Presentations, Curriculum Design

Signature Projects

GitHub Repository

Python, Gemini 1.5, FastAPI

Billy AI – Clinical Triage Assistant

2026

- Architected an AI-driven triage system for cyberbullying support, processing 500+ simulated scenarios
- Implemented trauma-informed design principles with localization for Nigerian youth demographics
- Achieved 92% accuracy in severity classification using few-shot prompting with Gemini 1.5 Pro

Live Demo

React, Node.js, MongoDB

Footprint Manager – Digital Privacy Platform

2026

- Developed a digital footprint analysis tool for UK citizens, emphasizing privacy-first infrastructure
- Engineered real-time risk assessment algorithms based on data exposure patterns
- Deployed on Vercel with CI/CD pipeline; currently serving 200+ active users in beta

Research Implementation

MATLAB, R, Stan

Weibull Analysis for Pipeline Infrastructure

2025

- Implemented Bayesian Weibull models for pipeline replacement policy optimization
- Presented findings at ICONDACOM International Conference (June 2025)
- Proposed model reduced maintenance costs by an estimated 18% in simulation studies

Publications & Conference Presentations

2026: (Under Review)

“Optimal Replacement Policies for Deteriorating Infrastructure Using Renewal-Reward Processes,” *Journal of Applied Probability* 2025: “Weibull-Modulated Delay Time Analysis for Pipeline Risk Assessment,” presented at ICONDACOM International Conference, Kano, Nigeria

2025: (Working Paper)

“Stochastic Modeling of Cyber Incident Triage: A Renewal Theory Approach to Security Operations”

Fellowships & Certifications

2026: Google DeepMind x Kaggle Hackathon Participant – Stochastic Executive Function track

2025: 3MTT Cybersecurity Programme (Cohort 3) – Federal Ministry of Communications, Innovation & Digital Economy, Nigeria

2025: ALX Alumni – Software Engineering Programme

2024: eHealth Africa Alumni – Software Engineering Programme

2024: Google Data Analytics Professional Certificate – Coursera

Professional Affiliations

- Nigerian Mathematical Society (NMS)
- Institute of Electrical and Electronics Engineers (IEEE) – Cybersecurity Special Interest Group
- Association for Computing Machinery (ACM)

Teaching Statement

“Bridge abstract mathematical theory with practical engineering applications, emphasizing computational thinking and security-aware design. Every equation should connect to a real-world system, and every system should be evaluated for resilience.”

Tools & Technologies

- **Programming:** Python, JavaScript/TypeScript, R, ○ **Frameworks:** React, Node.js, FastAPI, Express, MATLAB
- **Mathematics:** LaTeX, Stan, Jupyter, ○ **DevOps:** Git, GitHub Actions, Vercel, Docker, NumPy/SciPy
- **Design & Docs:** Markdown, Notion, Obsidian, ○ **Cybersecurity:** Wireshark, Nmap, OWASP guidelines, Figma